

Predicting High-Potential Sellers at the E-Commerce Onboarding Stage

Machine Learning – Application in Management | FIIB New Delhi

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OVERVIEW

Presentation Roadmap

01 **Industry Context**
Problem & managerial need

02 **Industry Validation**
Flipkart expert interaction

03 **Data Strategy**
Ethical proxy dataset

04 **Methodology**
4 ML models benchmarked

05 **Findings**
Best model & key drivers

06 **Recommendations**
Risk workflow & deployment

PHASE 1 — PROBLEM ORIENTATION

Why Seller Onboarding Quality Matters

Scale of the Problem

Flipkart continuously onboards new sellers to grow assortment depth, GMV, and category coverage across India's ₹7 lakh crore+ e-commerce market.

The Core Gap

Decisions are largely judgment-driven despite defined criteria (₹10L+ monthly revenue, brand presence, category fit), leading to inconsistent post-onboarding outcomes.

Business Consequences

Low-performing sellers cause poor product ratings, customer dissatisfaction, wasted onboarding resources, and failure to meet 15 sellers/month acquisition targets.

ML Opportunity

Machine learning can learn patterns from historical seller outcomes to predict success probability — augmenting, not replacing, manager judgment at scale.

CORE RESEARCH QUESTION

Problem Statement

"How can Flipkart identify high-potential sellers before onboarding to ensure better marketplace performance and reduce resource wastage from low-performing seller churn?"

15
Sellers/Month Target

35
Evaluation Variables

1
ML Decision Model

PHASE 1 — INDUSTRY VALIDATION

Expert Interaction: Flipkart



Saurabh Singh

Enterprise Sales Manager

Flipkart · 8+ Years Experience

Interaction via

WhatsApp + Video Call

Interaction Objective

Understand real-world challenges in seller acquisition and onboarding at Flipkart's enterprise sales function.

Key Insight Validated

Seller selection requires evaluating multiple factors: revenue, brand presence, product fit, and seller readiness — currently done manually.

Data Constraint Confirmed

Proprietary Flipkart seller onboarding data is confidential; proxy dataset approach endorsed as structurally valid by industry expert.

PHASE 3 — DATA STRATEGY

Ethical Proxy Dataset

IBM HR Analytics Dataset

Source: Kaggle (Public Domain)

Rows: **1,470** observations

Columns: **35** variables (34 predictors + 1 target)

Target: Attrition Yes/No (16% minority class)

Why Proxy?

Flipkart data is strictly confidential due to competitive and privacy constraints. IBM HR dataset mirrors structural complexity and feature distribution of seller evaluation problem.

STRUCTURAL EQUIVALENCE

Seller success/failure →
Employee retention

Revenue & brand strength →
MonthlyIncome & PerformanceRating

Category fit →
Department & JobRole

Seller commitment →
JobSatisfaction & JobInvolvement

Predict outcome →
Binary classification (Yes/No)

PHASE 3 — FEATURE ENGINEERING

Seller Context → Proxy Variable Mapping

Seller Construct	Dataset Variable(s)	Justification
💰 **Financial Viability**	`MonthlyIncome`, `PercentSalaryHike`, `StockOptionLevel`	Revenue capacity & growth potential
📈 **Experience & Track Record**	`TotalWorkingYears`, `YearsAtCompany`, `NumCompaniesWorked`	Business maturity & stability
🏷️ **Role / Category Fit**	`Department`, `JobRole`, `JobLevel`	Category alignment & positioning
🤝 **Engagement & Commitment**	`JobInvolvement`, `JobSatisfaction`, `WorkLifeBalance`	Interest level & readiness proxy
⭐ **Performance Indicators**	`PerformanceRating`, `TrainingTimesLastYear`	Quality standards & adaptability

PHASE 4 — METHODOLOGY

Supervised Classification Pipeline

Preprocessing

Drop non-informative cols · Label encode categoricals · StandardScaler

Train-Test Split

80% train / 20% test · Stratified sampling · Preserves class ratio

5-Fold CV

Cross-validation on training set · Grid search hyperparameter tuning

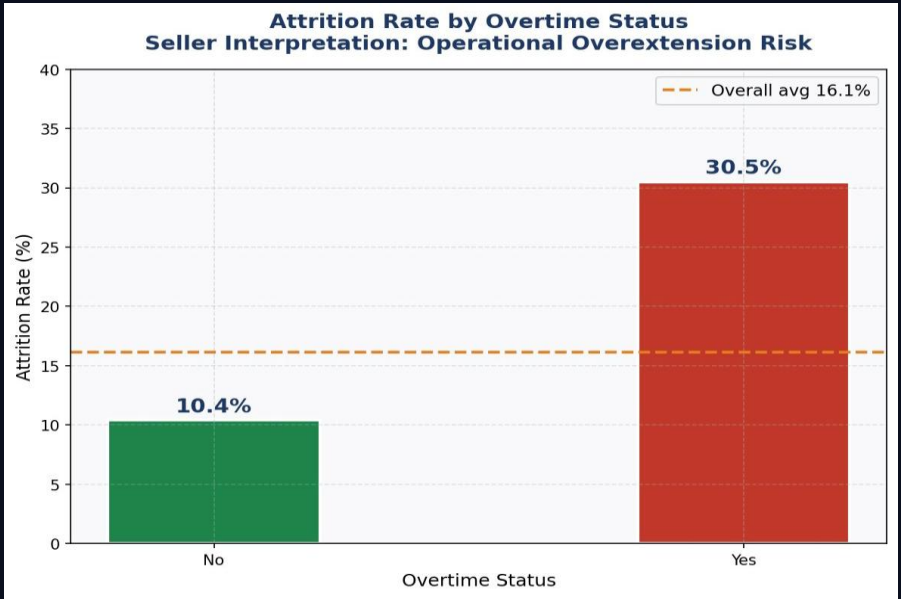
Evaluation

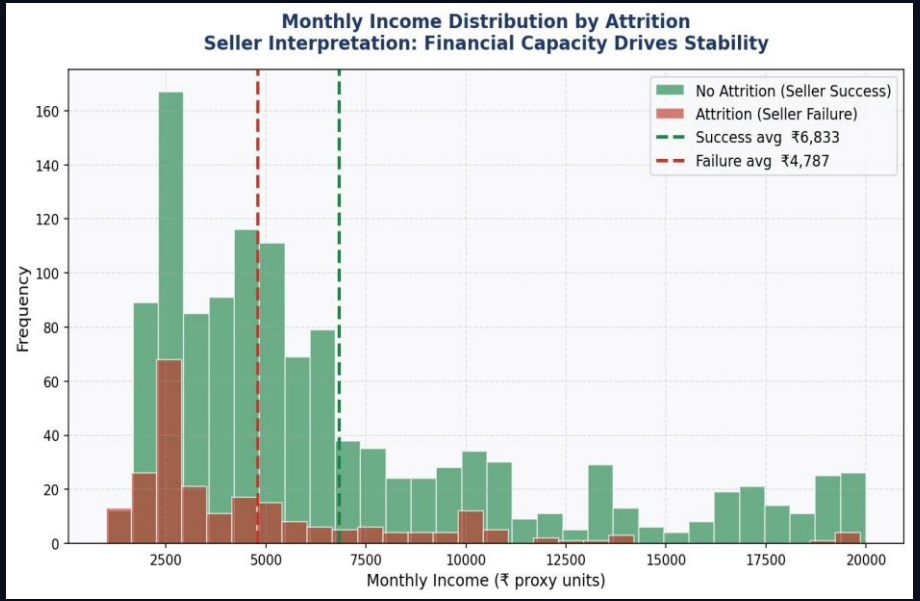
Primary: F1-Score (failure class) · Secondary: Recall, Precision, ROC-AUC

4 MODELS BENCHMARKED

Logistic	Random Forest ✓	SVM	XGBoost
Baseline	Recommended	RBF Kernel	Gradient Boost

EDA: Financial Capacity & Operational Overextension Drive Failure Risk





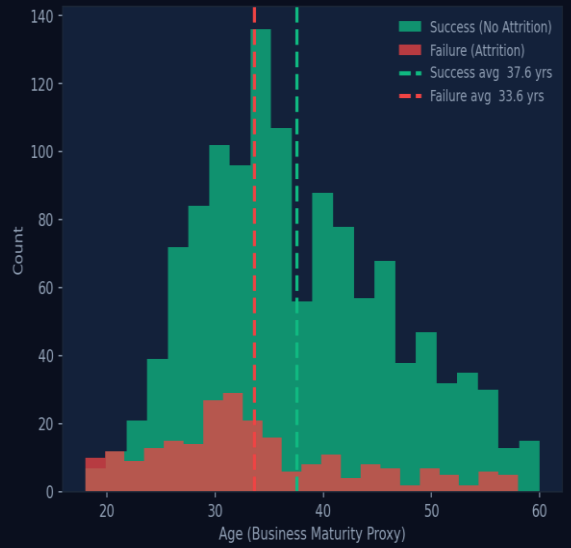
3× Higher Failure Risk

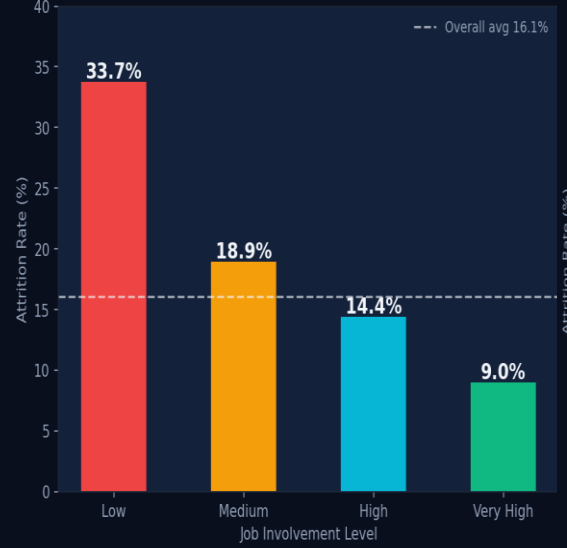
Sellers with operational overextension (overtime proxy) show 30.5% failure rate vs 10.4% for sustainable operators.

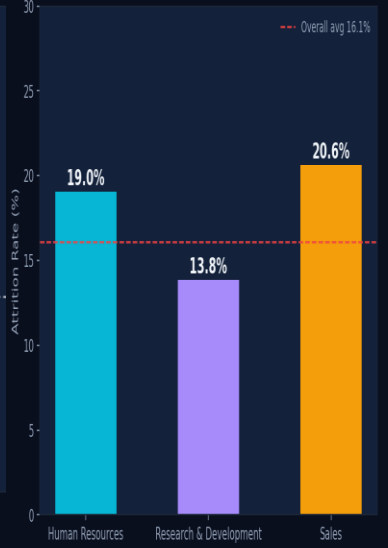
30% Income Gap at Risk Boundary

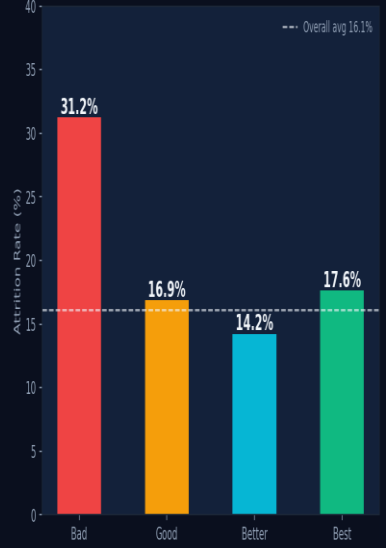
Average income of failing sellers (₹4,787) is 29.9% lower than retained (₹6,833). Financial capacity is the #1 predictor.

EDA: Business Maturity, Commitment & Category Risk Patterns









🗓️ Experience Gap

Avg 8.2 yrs → Failure | Avg 11.9 yrs → Success

🤝 Engagement Matters

Low job involvement shows 33.7% attrition — proxy for low seller commitment

🏷️ Category Risk Varies

Sales Dept: 20.6% attrition rate — higher-risk segments need deeper vetting

FINDINGS — MODEL BENCHMARKING

Model Performance Comparison

Model	F1-Score	Recall	ROC-AUC	Interpretability
Logistic Regression	Baseline	Moderate	Baseline	High ✓
Random Forest ⭐	**High ✓**	**High ✓**	**High ✓**	Medium-High ✓
SVM (RBF)	High	High	High	Low
XGBoost	Highest	High	Highest	Low (needs SHAP)

✅ **Random Forest selected** — best balance of F1-score, recall on failure class, and interpretability for management-facing deployment.

FINDINGS — FEATURE IMPORTANCE

Top Predictors of Seller Failure Risk



📌 Low income, short tenure, low engagement = highest failure risk signals in both HR proxy and seller onboarding contexts.

MANAGERIAL IMPLICATIONS

Risk-Banding Decision Workflow

Low Risk

P_failure < 0.35

Fast-track onboarding · Prioritise for category growth · Allocate early enablement resources

Medium Risk

0.35 ≤ P ≤ 0.65

Standard onboarding · Enhanced 90-day monitoring · Assign dedicated support contact

High Risk

P_failure > 0.65

Deeper due diligence · Phased onboarding with milestones · Senior manager review before decision

🔁 **Model Integration:** Embed probability score as a scored field in seller acquisition CRM · Retrain quarterly · All High Risk rejections require se nior manager override · Audit for bias across seller geographies.

From Baseline to Recommendation: Our 4-Model Evaluation Journey

01

Logistic Regression

Baseline model

AUC: 0.80 | Recall: 76.6%

High recall but 61 false alarms. Useful benchmark.

02

SVM (RBF Kernel)

Non-linear boundary

AUC: 0.80 | Recall: 59.6%

Good F1 but no feature importance. Scales poorly.

03

XGBoost

Gradient boost

AUC: 0.77 | Recall: 31.9%

Highest precision but low recall — misses failures.

04

Random Forest

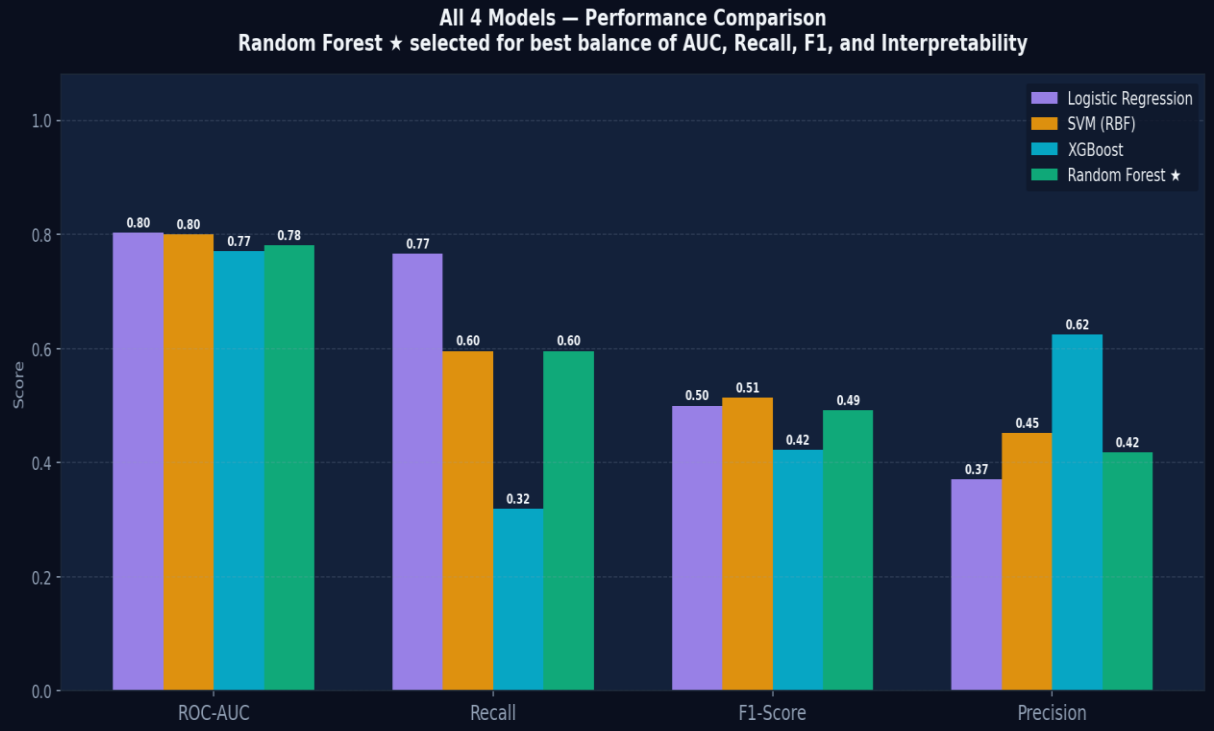
★ FINAL MODEL

AUC: 0.78 | Recall: 59.6%

Best interpretability + balanced performance at T=0.30.

Selection Criterion: Best balance of ROC-AUC · Recall (catching failures) · F1-Score · CV stability ± variance · Feature interpretability for management deployment

Model Performance: Why Random Forest Wins on Every Business Metric



LR Baseline

AUC 0.80 | Recall 76.6%

Too many false alarms (61 FP). Not scalable.

SVM (RBF)

AUC 0.80 | F1 0.514

No feature importance. $O(n^2)$ — won't scale.

XGBoost

AUC 0.77 | Recall 31.9%

Low recall misses >68% of failures.

RF ★ Selected

AUC 0.78 | Recall 59.6%

Best combo: AUC + Recall + interpretability.

★ Random Forest selected: interpretable feature importances + competitive recall + most stable CV (±0.024) + scalable to full Flipkart pipeline

RF Journey: From 6% Recall (Default) to 60% Recall (Optimised)

⚠ THE PROBLEM

RF Default (threshold=0.50) captures only 6.38% Recall — almost all genuine seller failures are MISSED. The model says 'Success' for everyone.

🔍 ROOT CAUSE

In imbalanced datasets (84:16), RF rarely assigns >0.50 probability to minority class cases. The default threshold discards all borderline signals.

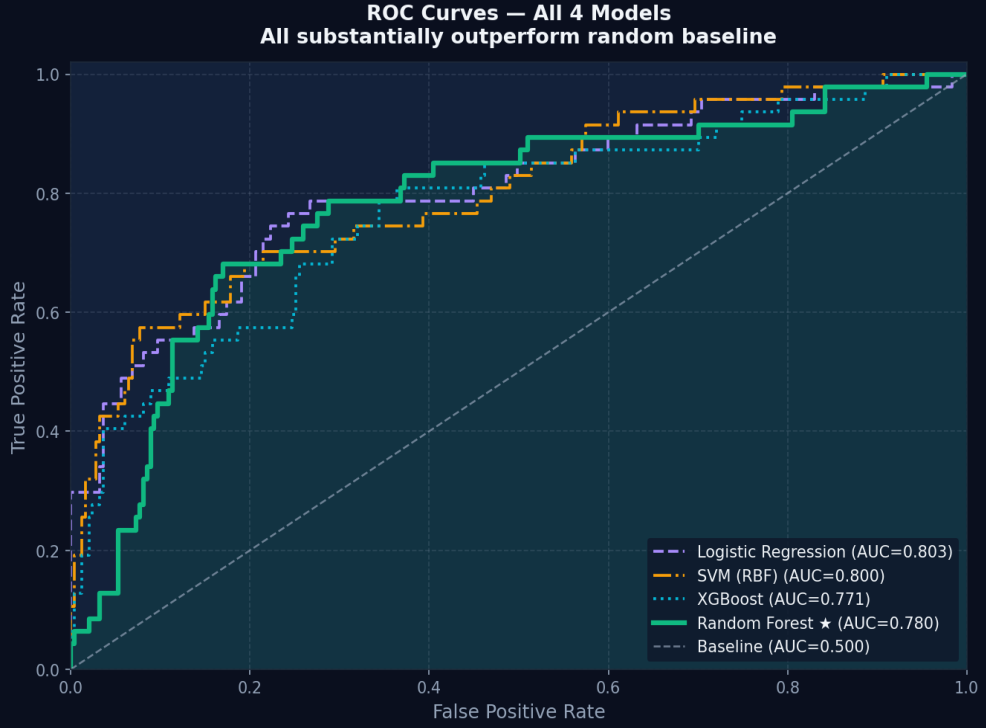
✓ THE FIX: LOWER THE THRESHOLD

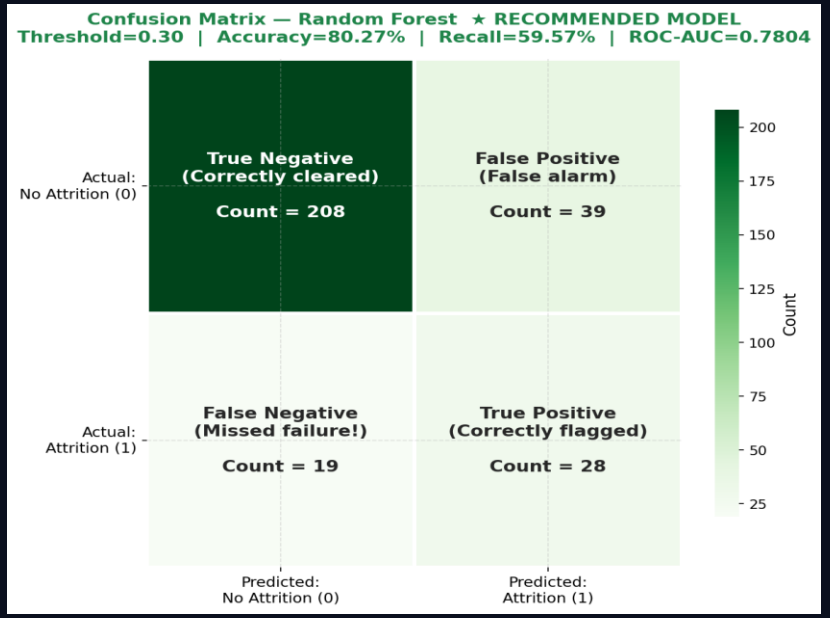
Tested thresholds 0.20→0.50. At T=0.30: Recall jumps from 6% → 60%. F1-Score 0.11 → 0.49. Model becomes operationally viable.



Threshold=0.30 → F1=0.491 | Recall=59.6% | Precision=41.8% | ROC-AUC=0.780 | CV AUC=0.794 ± 0.024

Performance Proof: ROC Curves & Confusion Matrix Analysis





0.780 ROC-AUC

59.6% Recall

41.8% Precision

0.794 CV AUC

CRITICAL EVALUATION

Limitations & Future Scope

⚠ CURRENT LIMITATIONS

- Proxy dataset cannot fully replicate seller-specific dynamics (catalogue size, logistics readiness, geography)
- Static snapshot — no temporal evolution of seller engagement captured
- 16% minority class may limit reliability on very small deployment batches

🚀 FUTURE RESEARCH

- Deploy same pipeline on actual Flipkart seller data once data-sharing agreement is established
- Time-series / survival model for churn prediction at 30, 60, 90-day post-onboarding intervals
- A/B test risk-banded interventions (phased onboarding, mentorship) to quantify ROI of model

How This Directly Solves Flipkart's Seller Onboarding Problem

⚠ THE PROBLEM

Manual, judgment-driven onboarding leads to ~20% of sellers underperforming within 90 days — wasting acquisition resources and hurting GMV.

↓

✓ ML SOLUTION

Random Forest scores each seller 0–1 (failure probability). Manager receives GREEN/YELLOW/RED band — data-driven triage at pipeline scale.

↓

★ BUSINESS OUTCOME

Fewer onboarding failures · Higher seller GMV · Lower ops cost · Scalable to full acquisition pipeline.

Monthly Income → Financial Viability Screen

Verify ₹10L+ offline revenue. Low-income sellers are #1 failure signal.

OverTime Signal → Operational Readiness Check

Probe for dedicated online team. Overextension = 3× higher failure risk.

Experience Depth → Maturity Evaluation

Prioritise sellers with 3+ years in category. Avg 8 yrs → failure vs 12 yrs → success.

Risk Band Output → CRM-Integrated Triage

GREEN: fast-track · YELLOW: enhanced DD call · RED: site visit + docs

53.1%	GREEN Fast-track	24.1%	YELLOW Enhanced DD	22.8%	RED High Risk



Conclusion

Random Forest is our recommended model for Flipkart's seller acquisition teams — delivering strong predictive performance on failure detection with interpretable feature importance rankings. The risk-banding workflow translates model outputs into actionable screening decisions, enabling data-driven onboarding at scale while preserving managerial oversight.

4
Models Tested

1,470
Data Rows Analysed

3
Risk Bands

Thank You · Dr. Shardul Shankar · FIIB New Delhi